- ENGINEERING MATERIALS -

- Dr. Nitesh Chindarkar

MODULE V : ADVANCED POLYMERS

(No. of hours – 03, Weightage of marks with option – 11)

- **Course Objective**: To study the different types of advanced polymers with their applications.
- **Course Outcome**: Recognize different advanced polymers for specific engineering applications.

MODULE V : ADVANCED POLYMERS

Sr. No.	TOPIC	REMARKS
1.	A) Conducting polymers	Basic concepts of conducting polymers, Types of conducting polymers: a) Intrinsic conducting polymers b) Extrinsically conducting polymers c) Doped Conducting Polymers d) Blended Conducting Polymers. Applications of Conductive Polymers.
2.	B) Biopolymers	Definition, Classification- Natural biopolymers (ie. Carbohydrates, Proteins, Nucleic acids) and Synthetic biopolymers (i.e. Microbial and Chemical synthesis), Advantages, Disadvantages and Applications.

Sr. No.	TOPIC	REMARKS
3.	C) Liquid crystal polymers	Definition, Types of LCP's (i.e. Thermotropic and Lyotropic LCP's), Structural Features of LCP's, Applications of LCP's.
4.	D) Intelligent (smart) polymers	Types of Smart Polymers: Thermoresponsive Polymers, pH-Responsive Polymers, Light -Responsive (Photo-Responsive) Polymers, Applications in various fields.

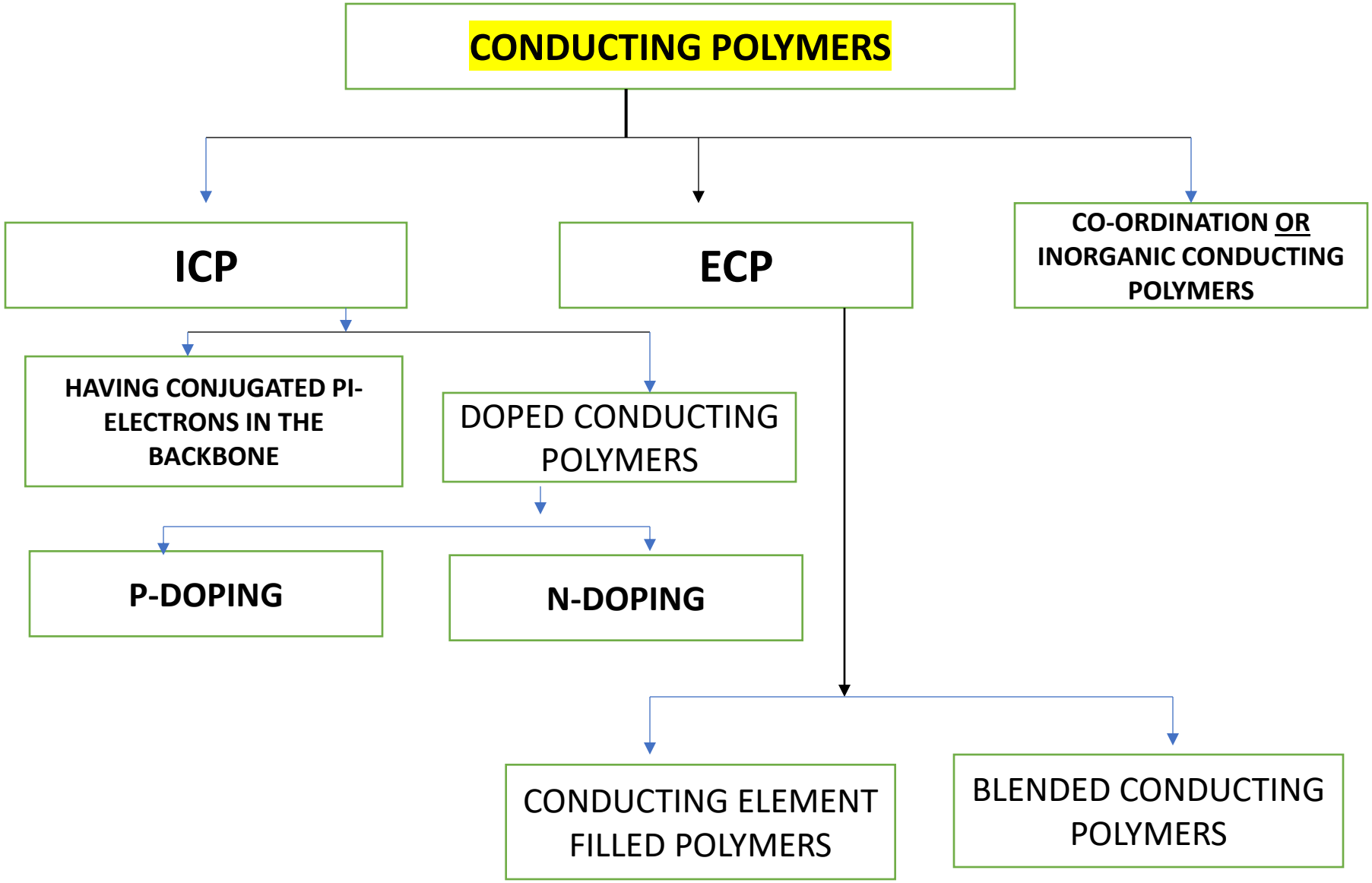
Reference Books (Advanced Polymers):

1. Polymer Chemistry: The basic concept and application by Dr. Rohit Kumar Bargah, BFC Publications

2. Smart Polymers and their Applications by Maria Rosa Aquillar, Woodhead Publishing in Materials.

3. Liquid Crystalline Polymers: Synthesis, Properties and Applications (Polymer Science) by Vikas Mittal, Central West Publishing.

Conducting Polymers Types



Conducting Polymers

- Polymers may also have conductivities corresponding to those of semiconductors.
- Two conditions for a polymer to become conducting are:

  - They possess conjugated double bonds.

  - Polymer has to be disturbed either by removing or adding electron to the material. This process is called as doping.

Intrinsically Conducting Polymer (ICP)

- This is a polymer whose backbone or associated groups consist of delocalised electron pair or residual charge.
- Such polymers essentially contain conjugated pi-electrons backbone, which is responsible for electrical charge.
- In an electric field, conjugated pi-electrons of the polymer gets excited and thereby can be transported through the solid polymeric material.

e.g. Polyquinoline, polyaniline, polyanthracene, etc.

Doped conducting polymer

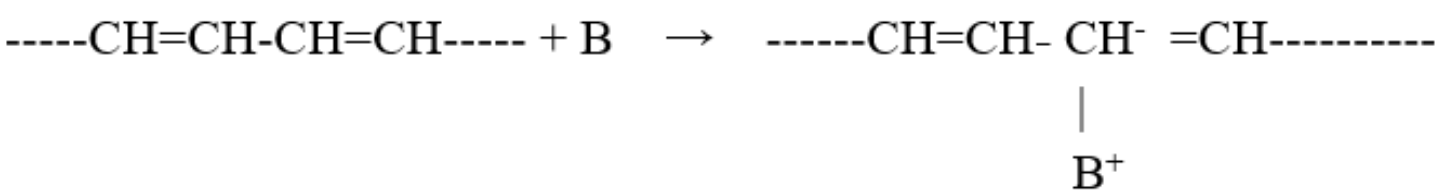
- This is obtained by exposing a polymer to a charged transfer agent in either gas or in solution.
- Consequently, the conductivity of ICP can be increased by creating either positive or negative charges on the polymer backbone by oxidation or reduction.
- This technique is called doping, is of **two** types:

a) p-doping involves treating an ICP with a Lewis acid, thereby the oxidation process takes place and positive charge on the backbone is created. Some of the common p-dopant used are I_2, Br_2, $FeCl_3$, etc.

$$(C_2H_2)n + 2\ FeCl_3 \rightarrow (C_2H_2)^{+}FeCl_4^{-} + FeCl_2$$

Doped conducting polymer

a) n-doping involves treating an ICP with a Lewis base; thereby the reduction takes place and negative charge on the polymer backbone are created. Some of the n-dopant are Li, Na, Ca, $FeCl_3$, etc.



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Extrinsically conducting polymers

- These are those polymers whose conductivity is due to the presence of “externally” added ingredients in them
- Two types:

a) Conducting element-filled polymer:

- Its a polymer with conducting element such as carbon black, metallic fibres, metal oxides, etc.

- Generally carbon black is used on account of its: very high surface area, high porosity and its filamentous properties.

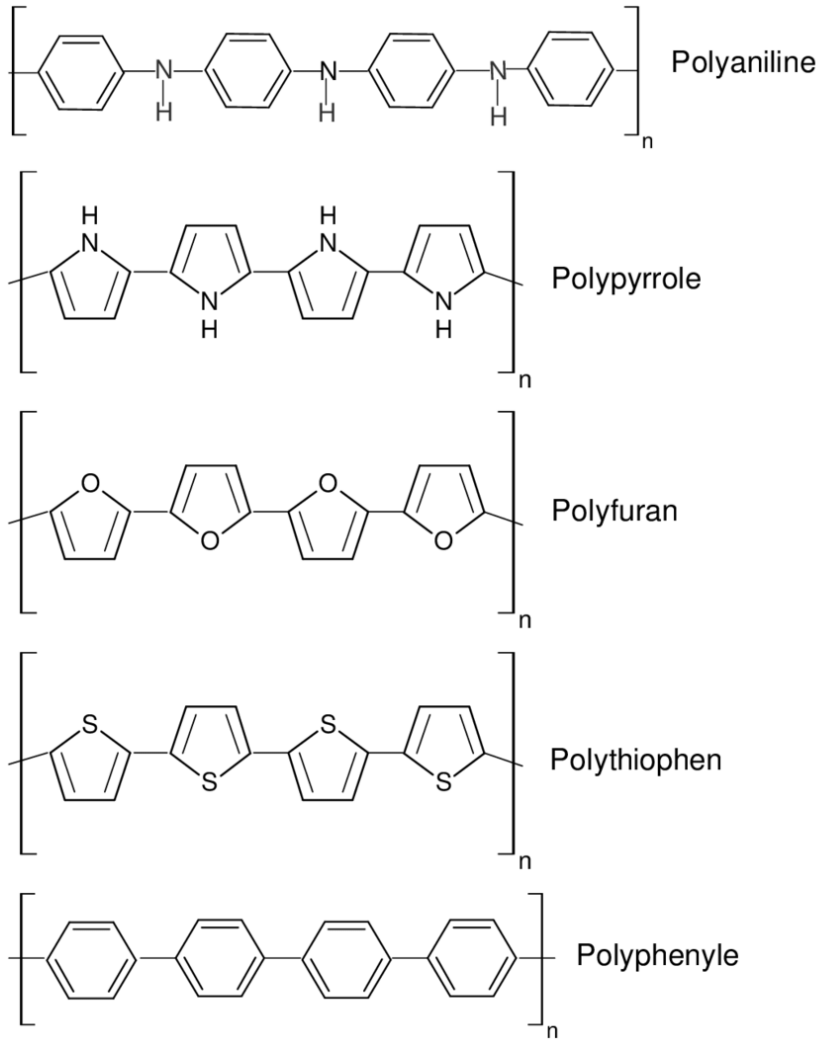
Extrinsically conducting polymers

b) Blended conducting polymer:

- Its a product obtained by blending a conventional polymer with conducting polymer either by chemical or physical change.

- Such polymer can be easily processed and possess better physical, chemical and mechanical properties.

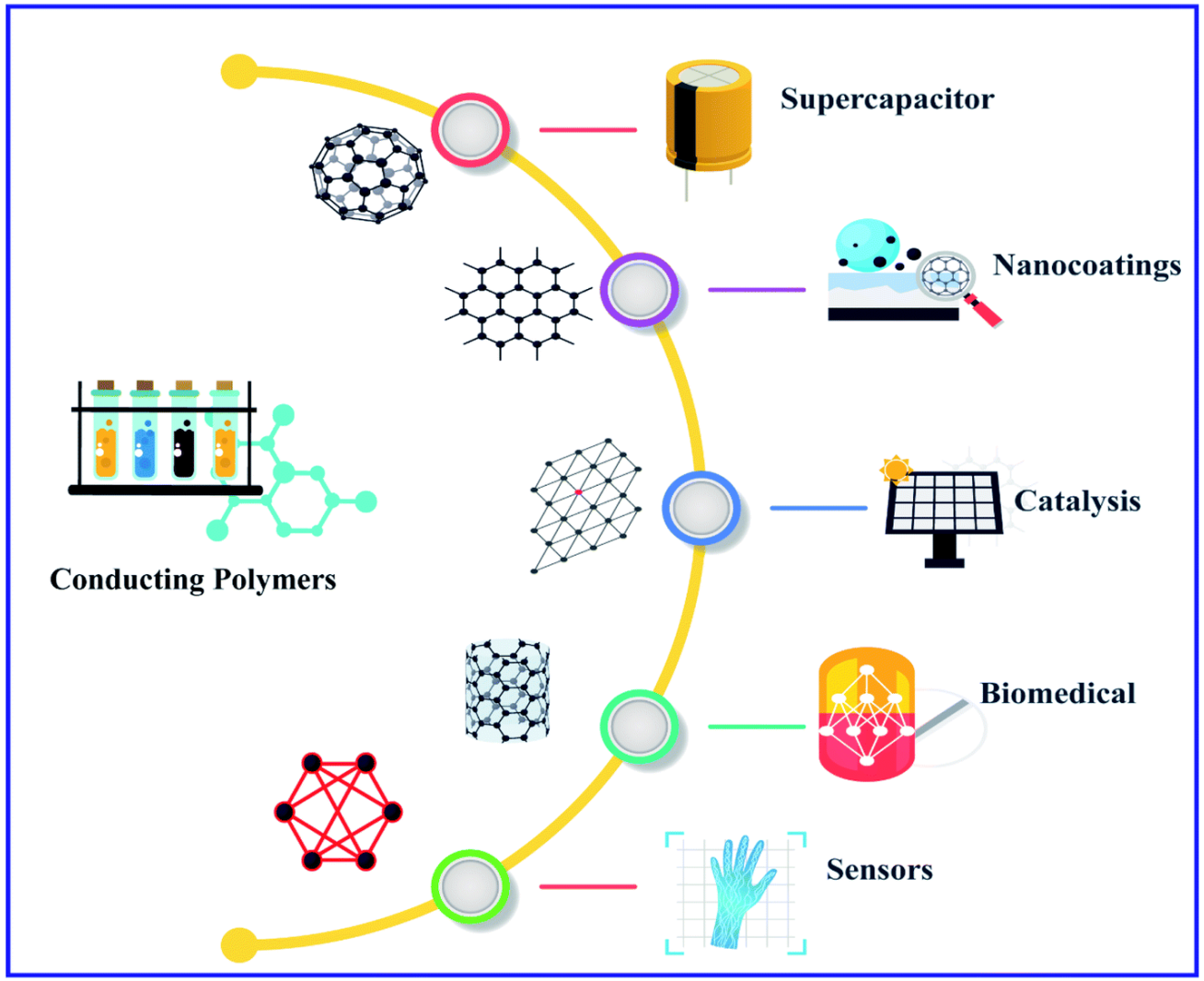
Examples of Conducting Polymers



Applications of conducting polymers

1. **Batteries**: Conducting polymers are used as electrodes in rechargeable batteries, improving their performance and lifespan.
2. **Organic Light-Emitting Diodes (OLEDs)**: Conducting polymers are used as hole-injecting materials in OLEDs, improving their brightness and lifespan.
3. **Sensors**: Conducting polymers are used in various sensors, such as gas sensors, biosensors, and pressure sensors.
4. **Tissue Engineering**: Conducting polymers are used as scaffolds for tissue engineering, providing electrical stimulation for cell growth.
5. **Corrosion Protection**: Conducting polymers are used as coatings for corrosion protection of metals.
6. **Water Treatment**: They are used in water treatment technologies, such as electrochemical water treatment.

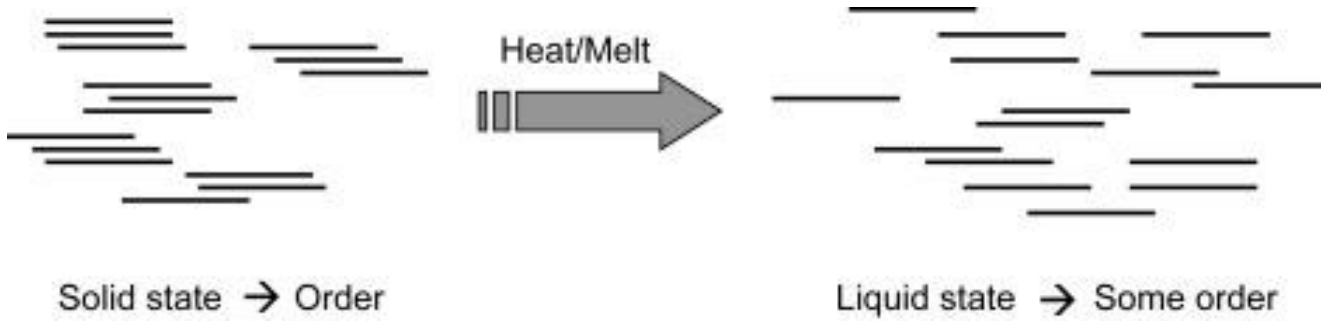
Applications of conducting polymers



Liquid Crystalline Polymers

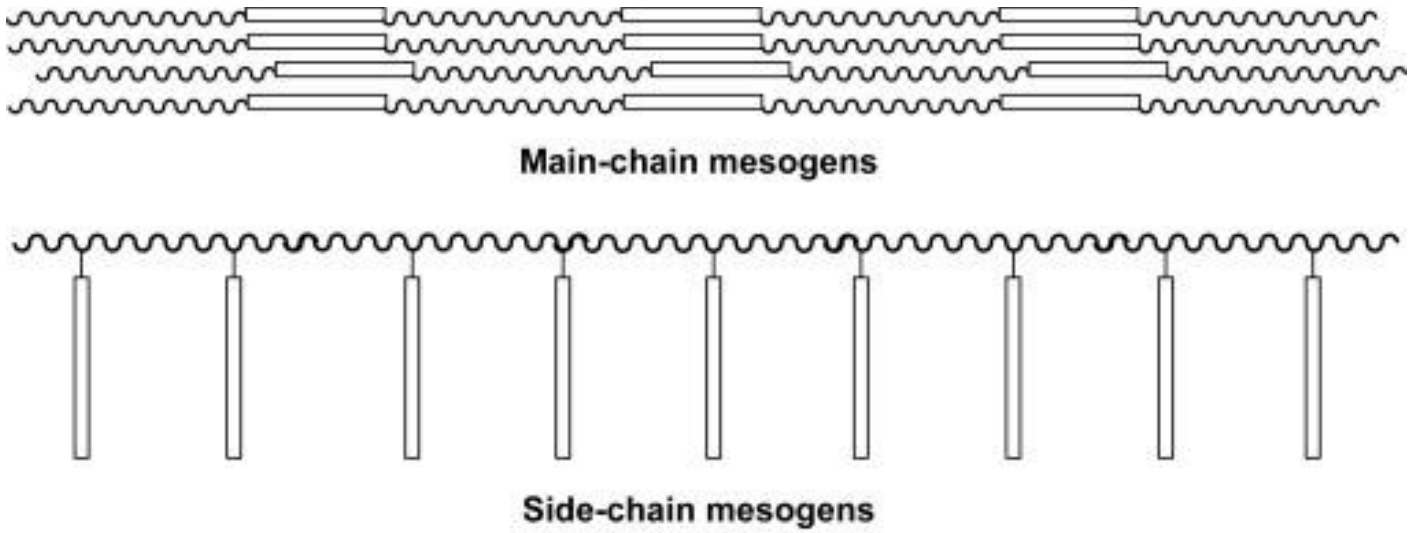
Introduction:

- Liquid crystalline polymers (LCPs) are a special type of thermoplastic that exhibits properties between highly ordered solid crystalline materials and amorphous disordered liquids over a well-defined temperature range.
- The term liquid crystal originates from the fact that the crystal structure of the polymer is partly maintained in the liquid phase above the melting point.



Liquid Crystalline Polymers

- This is because the polymer consists of ordered, rigid rods, or mesogens either along the polymer backbone or on the polymer's side chains (Fig.).



- Thus, LCPs exhibit a highly ordered structure in both the melt and solid states. This results in their properties being in between those of liquids and crystalline solids. **For example, they can have liquid-like flow properties and solid-like tensile strength and stiffness.**

Structural Features of Liquid Crystalline Polymers

- Liquid Crystalline Polymers (LCPs) have unique structural features that enable their exceptional properties. Here are some key structural features of LCPs:

1. Mesogenic Groups- LCPs contain mesogenic groups, which are rigid, rod-like structures that exhibit liquid crystalline behavior. These groups can be composed of aromatic rings, heterocyclic rings, or other rigid structures.

2. Flexible Spacers- Flexible spacers, such as alkyl chains or ether linkages, connect the mesogenic groups. These spacers provide flexibility and help to reduce the melting point of the polymer.

Structural Features of Liquid Crystalline Polymers

3. Main Chain or Side Chain Architecture- LCPs can have either a main chain or side chain architecture. Main chain LCPs have the mesogenic groups incorporated into the polymer backbone. Side chain LCPs have the mesogenic groups attached to the polymer backbone as side chains.

4. Molecular Weight and Polydispersity- LCPs typically have high molecular weights, which contribute to their exceptional mechanical properties. The polydispersity index (PDI) of LCPs can affect their processing and properties.

5. Crystallinity and Liquid Crystallinity- LCPs can exhibit crystallinity, liquid crystallinity, or a combination of both. The degree of crystallinity and liquid crystallinity can impact the properties of LCPs.

Structural Features of Liquid Crystalline Polymers

These structural features enable LCPs to exhibit unique properties, such as:

- **High mechanical strength and stiffness**
- **High thermal stability**
- **Low melt viscosity**
- **Anisotropic properties**
- **Self-reinforcement**

Liquid Crystalline Polymers

- LCPs may be divided into two main categories, depending on how they achieve their liquid-crystalline characteristics.
- Lyotropic LCPs are obtained from solutions; that is, when polymers are dissolved in solvents. E.g. Poly(p-phenylene terephthalamide) (PPTA), Poly(2,5-benzoxazole) (ABPBO).
- Thermotropic LCPs are produced via heat; that is, when polymers are heated to a molten state. They are more commercially viable and are the materials used in medical device applications. E.g. Polyesters, Polyamides, Polyimides.

Applications of Liquid Crystalline Polymers

1. **High-Performance Composites**

- LCPs are used as matrices in high-performance composites for aerospace, automotive, and sports equipment applications.

- They provide exceptional mechanical properties, thermal stability, and resistance to chemicals and abrasion.

2. **Electrical and Electronic Components**

- LCPs are used in electrical and electronic components, such as connectors, sockets, and switches.

- They provide high thermal stability, chemical resistance, and electrical insulation.

Applications of Liquid Crystalline Polymers

3. **Medical Devices and Implants**

- LCPs are used in medical devices and implants, such as surgical instruments, catheters, and bone screws.

- They provide biocompatibility, high mechanical strength, and resistance to sterilization methods.

4. **Packaging and Barrier Films**

- LCPs are used in packaging and barrier films for food, pharmaceutical, and cosmetic applications.

- They provide high barrier properties, thermal stability, and chemical resistance.

Applications of Liquid Crystalline Polymers

5. **Membranes and Separations**

- LCPs are used in membranes and separations, such as filtration, separation, and purification applications.

- They provide high selectivity, thermal stability, and chemical resistance.

6. **3D Printing and Additive Manufacturing**

- LCPs are used in 3D printing and additive manufacturing, providing high thermal stability, mechanical strength, and chemical resistance.

Applications of Liquid Crystalline Polymers

7. **Optical Components and Displays**

- LCPs are used in optical components and displays, such as lenses, prisms, and liquid crystal displays (LCDs).

- They provide high optical clarity, thermal stability, and chemical resistance

Biopolymers

Definition:

‘Biopolymers’ are naturally occurring long-chain molecules which are involved in biological changes important for our lives.

e.g. polysaccharides (carbohydrates), lipids (fats), proteins, nucleic acids (DNA, RNA), and natural rubber. These polymers are present in living matter (animals and plants).

Natural Biopolymers

- CARBOHYDRATES:

1. General formula- $C_x(H_2O)_y$
2. Carbohydrates are divided into Monosaccharides (e.g. ribose, glucose), oligosaccharides (maltose, sucrose) and polysaccharides (starch, cellulose, glycogen) with general formula $(C_6H_{10}O_5)_n$ where n=1000-3000.

Functions:

1. They form **structural components of cells**. For example, cellulose is present in the cell walls of all plant cells.
2. They function **as bio-fuels** to provide energy for the functioning of living system and doing work.

CARBOHYDRATES

- Functions:

3. They act **as storage of chemical energy** in the form of glycogen in liver. In case of emergency (such as illness or fasting), glycogen hydrolysis to glucose, which then supplies the necessary energy. Similarly, starch is the main energy storage polysaccharide in plants.

4. Cellulose present in grass and plants acts **as food** for various grazing animals because they contain in their body specific enzymes which can hydrolyze cellulose into glucose next point.

5. Some carbohydrates **acts as food** e.g. cane sugar, glucose and starch.

Natural Biopolymers

- PROTEINS:

1. Proteins are nitrogenous organic compounds of high molecular masses, synthesized by plants and animals, which on hydrolysis (by enzymes) yield amino acids.
2. Thus, proteins are polymeric amides, composed of long chains of numerous amino acids, linked by **peptide bonds**. Chemically, proteins yield alpha-amino acids on hydrolysis.

Functions:

1. They serve **as fuel to yield energy**.
2. They serve **as the structural component** of the living organisms.

PROTEINS

- Functions:

3. They are responsible for the **functional characteristics**.

4. They **serve as enzymes**, which catalyze biochemical reactions.

5. They help in the formation of haemoglobin, chromosomes, etc.

6. They form hormones and antibodies (protective agent), which are highly essential for animals.

7. They act as mode of transport e.g. haemoglobin (present in blood) acts as carrier of oxygen and carbon dioxide.

8. They take part in genetics.

NUCLEIC ACIDS

Nucleic acids polynucleotides (a nucleotide consist of sugar, a phosphate group and a nitrogen base residue). Thus, nucleic acids are high polymers of nucleotide. In other words, nucleotides are the repeating are the repeating of nucleic acids.

Structure of nucleic acid: It consists of three parts, namely, (i) nitrogen-containing base, (ii) pentose sugar, (iii) phosphate group.

Types of nucleic acid:

(i) DNA: It is responsible for the inheritance of genetic character of a particular species of living cell.

(ii) RNA: r-RNA – This forms the site for protein synthesis. m-RNA – It carries the genetic information from DNA to ribosomes, where protein is synthesized. t-RNA: It acts as carriers of amino acids.

SYNTHETIC BIOPOLYMERS

- Production methods:

1. Microbial Synthesis: Microorganisms such as bacteria and yeast are used to produce biopolymers through fermentation. Examples include polyhydroxyalkanoates (PHA) and poly(lactic acid) (PLA).
2. Chemical Synthesis: Chemical reactions are used to produce biopolymers from renewable resources such as starch, cellulose, and proteins. Examples include polylactic acid (PLA) and polybutylene succinate (PBS).

SYNTHETIC BIOPOLYMERS

- Advantages:

1. **Biodegradable**: Synthetic biopolymers are biodegradable, reducing plastic waste and environmental pollution.
2. **Renewable Resources**: Many synthetic biopolymers are produced from renewable resources, reducing dependence on fossil fuels.
3. **Customizable**: Synthetic biopolymers can be tailored to have specific properties, such as tensile strength, elasticity, and biodegradability.
4. **Competitive Pricing**: Synthetic biopolymers can be produced at competitive prices, making them a viable alternative to traditional plastics.

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SYNTHETIC BIOPOLYMERS

- Disadvantages:

1. **Higher Production Costs**: While prices are decreasing, synthetic biopolymers can still be more expensive to produce than traditional plastics.
2. **Limited Availability**: Synthetic biopolymers are not yet widely available, limiting their adoption in various industries.
3. **Performance Limitations**: Synthetic biopolymers may not have the same performance characteristics as traditional plastics, such as durability and shelf life.
4. **End-of-Life Management**: While biodegradable, synthetic biopolymers require specific end-of-life management practices to ensure proper decomposition.

SYNTHETIC BIOPOLYMERS

- Applications:

1. **Packaging**: Synthetic biopolymers are used in packaging applications, such as food containers, bags, and wraps.
2. **Textiles**: Synthetic biopolymers are used in textile applications, such as clothing, carpets, and upholstery.
3. **Medical Devices**: Synthetic biopolymers are used in medical device applications, such as implants, sutures, and wound dressings.
4. **Agriculture**: Synthetic biopolymers are used in agricultural applications, such as mulch films, greenhouse films, and biodegradable pesticides.
5. **Consumer Goods**: Synthetic biopolymers are used in various consumer goods, such as disposable cutlery, straws, and bags.

Intelligent (smart) polymers

- Smart polymers are a class of advanced polymers that exhibit a change in their properties in response to external stimuli.
- These stimuli can include electrical, mechanical (stress or strain), thermal, light, magnetic, biological, or chemical influences.

Types of Intelligent (smart) polymers

1. Thermoresponsive Polymers,
2. pH-Responsive Polymers,
3. Light -Responsive (Photo-Responsive) Polymers

Thermoresponsive Polymers

- Thermoresponsive polymers are a class of smart materials that change their properties or behavior in response to changes in temperature. These polymers are also known as temperature-sensitive or thermo-responsive polymers.
- Some common characteristics of thermoresponsive polymers include:

1. Reversible phase transitions
2. Temperature-dependent solubility
3. Changes in shape, size, or color

Types of Thermoresponsive Polymers

1. Lower Critical Solution Temperature (LCST) Polymers: These polymers are soluble in water below a certain temperature (LCST) and become insoluble above it. Examples include PNIPAM and PVME.
2. Upper Critical Solution Temperature (UCST) Polymers: These polymers are insoluble in water below a certain temperature (UCST) and become soluble above it. Examples include poly(acrylic acid) and poly(methacrylic acid).

Thermoresponsive polymers

- Examples:

1. Poly(N-isopropylacrylamide) (PNIPAM)
2. Poly(vinyl methyl ether) (PVME)
3. Poly(ethylene oxide) (PEO)

- Applications:

1. Biomedical devices (e.g., drug delivery, tissue engineering)
2. Sensors and actuators
3. Smart textiles and fabrics
4. Separation and purification technologies

pH-responsive polymers

- pH-responsive polymers are a class of smart materials that change their properties or behavior in response to changes in pH levels. These polymers are also known as pH-sensitive or acid-responsive polymers.
- Examples of pH-Responsive Polymers:

1. **Poly(acrylic acid) (PAA)**: PAA is a pH-responsive polymer that contains carboxylic acid groups

2. **Poly(methacrylic acid) (PMAA)**: PMAA is a pH-responsive polymer that contains carboxylic acid groups

Properties of pH-responsive polymers

1. pH-Dependent Solubility: pH-responsive polymers can change their solubility in response to changes in pH levels.
2. Conformational Changes: pH-responsive polymers can undergo conformational changes, such as unfolding or folding, in response to changes in pH levels.
3. Changes in Swelling Behavior: pH-responsive polymers can change their swelling behavior in response to changes in pH levels.

Types of pH-Responsive Polymers

1. Weak Polyelectrolytes: Weak polyelectrolytes are pH-responsive polymers that contain **weakly** acidic or basic groups, such as carboxylic acid or amine groups.
2. Strong Polyelectrolytes: Strong polyelectrolytes are pH-responsive polymers that contain **strongly** acidic or basic groups, such as sulfonic acid or quaternary ammonium groups.

Applications of pH-Responsive Polymers

1. **Drug Delivery Systems**: pH-responsive polymers are being explored for use in drug delivery systems, where they can release drugs in response to changes in pH levels.
2. **Biomedical Devices**: pH-responsive polymers are being used to develop biomedical devices, such as biosensors and implantable devices.
3. **Tissue Engineering**: pH-responsive polymers are being used to develop tissue engineering scaffolds that can respond to changes in pH levels.
4. **Environmental Applications**: pH-responsive polymers are being explored for use in environmental applications, such as water treatment and pollution remediation.

Light-responsive (photo-responsive) polymers

- Light-responsive (photo-responsive) polymers are a class of smart materials that change their properties or behavior in response to light. These polymers are also known as photo-reactive or photo-sensitive polymers.

Examples of Light-responsive Polymers:

1. **Poly(ethylene glycol) (PEG) Derivatives**: PEG derivatives can be made photo-responsive by incorporating photo-cleavable or photo-cross-linkable groups.
2. **Poly(acrylamide) (PAM) Derivatives**: PAM derivatives can be made photo-responsive by incorporating photo-cleavable or photo-cross-linkable groups.
3. **Spiropyran-Containing Polymers**: Spiropyran-containing polymers are photo-chromic polymers that change color in response to light.

Properties of Light-responsive Polymers

1. Photo-Induced Conformational Changes: Light-responsive polymers can undergo conformational changes, such as folding or unfolding, in response to light.
2. Photo-Induced Solubility Changes: Light-responsive polymers can change their solubility in response to light.
3. Photo-Induced Cross-Linking: Light-responsive polymers can undergo cross-linking reactions in response to light.

Types of Light-Responsive Polymers

1. Photo-Cleavable Polymers: Photo-cleavable polymers contain groups that can be cleaved by light, leading to changes in the polymer's properties.
2. Photo-Cross-Linkable Polymers: Photo-cross-linkable polymers contain groups that can be cross-linked by light, leading to changes in the polymer's properties.
3. Photo-Chromic Polymers: Photo-chromic polymers change color in response to light.

Applications of Light-Responsive Polymers

1. **Optical Data Storage**: Light-responsive polymers are being explored for use in optical data storage applications.
2. **Biomedical Devices**: Light-responsive polymers are being used to develop biomedical devices, such as photo-controlled drug delivery systems.
3. **Soft Robotics**: Light-responsive polymers are being used to develop soft robots that can change shape in response to light.
4. **Smart Coatings**: Light-responsive polymers are being used to develop smart coatings that can change color or properties in response to light.


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